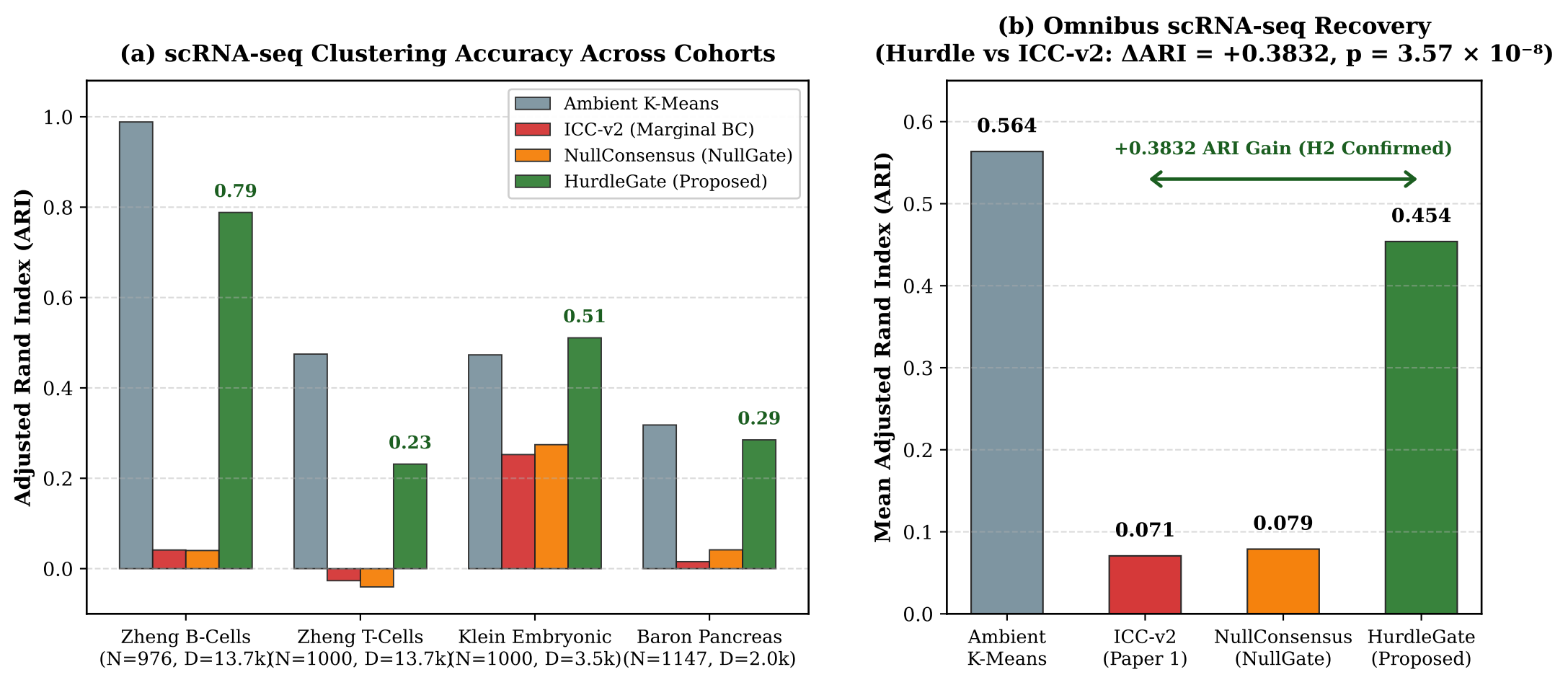
(a) scRNA-seq Clustering Accuracy Across Cohorts
Adjusted Rand Index (ARI)
Ambient K-Means
ICC-v2 (Marginal BC)
NullConsensus (NullGate)
HurdleGate (Proposed)
0.79
0.23
0.51
0.29
Zheng B-Cells (N=976, D=13.7k)
Zheng T-Cells (N=1000, D=13.7k)
Klein Embryonic (N=1000, D=3.5k)
Baron Pancreas (N=1147, D=2.0k)
(b) Omnibus scRNA-seq Recovery
(Hurdle vs ICC-v2: ΔARI = +0.3832, p = 3.57 × 10⁻⁸)
Mean Adjusted Rand Index (ARI)
0.564
+0.3832 ARI Gain (H2 Confirmed)
0.454
0.071
0.079
Ambient K-Means
ICC-v2 (Paper 1)
NullConsensus (NullGate)
HurdleGate (Proposed)